VICTORYAI

OUD RECOVERY COMPANION · OUTCOMES ANALYTICS PLATFORM

Clinical Evidence *Dossier*

The Peer-Reviewed Evidence Base for Each Therapeutic Mechanism Implemented in the VictoryAI Platform

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STATEMENT OF SCIENTIFIC INTEGRITY

How to Read This Document

VictoryAI has not yet completed its own clinical pilot. This dossier therefore does not present efficacy data for the VictoryAI platform itself. To do so would be scientifically improper and would undermine the very partnership we are proposing. Instead, this dossier presents the peer-reviewed evidence base for each individual therapeutic mechanism that VictoryAI has implemented, drawn from randomized controlled trials, FDA submissions, systematic reviews, and meta-analyses published in journals including the American Journal of Psychiatry, JAMA Psychiatry, JAMA Network Open, Drug and Alcohol Dependence, and Advances in Therapy.

The thesis of this document is straightforward. Every component of the VictoryAI platform — digital cognitive behavioral therapy, contingency-management-style reinforcement, family-based engagement, remote therapeutic monitoring, voice biomarkers for early relapse detection, and structured medication-adherence support — corresponds to an intervention modality with substantial published evidence. VictoryAI's contribution is not to invent a new therapeutic mechanism. It is to integrate six independently validated mechanisms into a single, mobile-delivered, EMR-integrated platform purpose-built for the operational reality of large MAT providers like BayMark.

The proposed BayMark joint innovation pilot is the path by which VictoryAI generates its own platform-specific clinical evidence. Section 5 of this dossier describes the IRB-ready study protocol that would govern that pilot. The protocol is designed to produce data that meets the methodological standards of the journals named above, and the publication and dissemination plan in Section 7 commits to peer-reviewed reporting of the results — favorable, unfavorable, or null.

THREE THINGS THIS DOSSIER DOES NOT DO

First, it does not claim that VictoryAI has demonstrated clinical efficacy. It has not yet been tested. Second, it does not claim that the published evidence for individual therapeutic mechanisms automatically transfers to the VictoryAI implementation of those mechanisms — that is precisely what the proposed pilot will test. Third, it does not present any statistic, effect size, or outcome attributed to VictoryAI itself. Every quantitative claim in this document is sourced to a published study of a different intervention, and every such claim is cited.

The reader should evaluate this dossier as the answer to one question: is the underlying science strong enough to justify a paid pilot at three to five California BAART clinics that would generate the platform-specific evidence VictoryAI currently lacks? The dossier is structured to support a reasoned yes or no to that question — not to manufacture a yes through overclaiming.

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EXECUTIVE SUMMARY

Six Mechanisms, One Platform

Opioid use disorder is the most lethal psychiatric condition in the United States. More than 80,000 Americans died from opioid-involved overdose in 2023. Medication for opioid use disorder (MOUD) — buprenorphine, methadone, or naltrexone — is the single most effective intervention available, yet across published trials, retention on buprenorphine sits at approximately fifty percent or less at six months.

This dossier reviews the evidence for six therapeutic mechanisms that have been shown, in randomized controlled trials and meta-analyses, to improve outcomes for patients on MOUD. Each mechanism is implemented in the VictoryAI platform. The mechanisms are: *digital cognitive behavioral therapy* (Carroll and colleagues, 2008–2024), *contingency management* (Bolívar et al., 2021, JAMA Psychiatry meta-analysis of 74 RCTs), *family-based engagement via CRAFT* (Meyers et al., 1999; Brigham et al., 2014), *remote therapeutic monitoring with structured engagement data capture*, *voice biomarkers for early mood and relapse detection* (multiple smartphone-based studies, 2020–2025), and *structured MAT adherence support* with real-time clinician dashboards. Each is documented in Section 2 with specific citations.

The most directly comparable platform is reSET-O, the prescription digital therapeutic developed by Pear Therapeutics and granted FDA market authorization in December 2018. reSET-O implements digital CBT plus contingency management for patients on buprenorphine. The pivotal NIDA-sponsored trial of reSET-O randomized 170 patients with OUD over 12 weeks and demonstrated significantly improved retention compared with treatment as usual. Subsequent real-world claims-based analyses of more than 1,800 reSET-O patients showed reduced healthcare resource utilization, fewer inpatient stays and emergency department visits, and improved buprenorphine adherence at 12 months relative to controls.

VictoryAI's hypothesis is that combining all six mechanisms into one integrated platform — with the structured engagement data flowing into a real-time provider analytics dashboard and an EMR-billable code export — will produce outcomes that improve on the published reSET-O effect sizes for the BayMark population. That hypothesis has not been tested. The proposed pilot would test it.

What the Dossier Asks BayMark to Conclude

The dossier asks BayMark's Chief Medical Officer and clinical leadership to conclude four things, and only four:

1. The therapeutic mechanisms VictoryAI implements are individually supported by published, peer-reviewed evidence of meaningful effect sizes in patients with substance use disorder, including OUD specifically.

2. The integration of these mechanisms into a single mobile platform with EMR integration is novel and warrants empirical evaluation.
3. The proposed clinical study protocol in Section 5 is methodologically sound, IRB-ready, and capable of generating publication-quality outcomes data.
4. The risk profile of the platform is acceptable, the patient safety architecture is sufficient, and the limitations described in Section 8 are appropriate to a pre-pilot stage of development.

If BayMark's clinical leadership is comfortable affirming those four conclusions, the joint innovation pilot moves forward. If not, this dossier identifies the specific objections that VictoryAI must address before a pilot is justified.

SECTION 01

The Clinical Problem

Any clinical evidence dossier must begin by establishing the magnitude and unmet need of the problem the intervention is designed to address. For BayMark, this is largely familiar territory — the organization treats more OUD patients than any other provider in the United States and Canada. This section restates the problem in terms relevant to the specific clinical and operational decisions BayMark's leadership will make about VictoryAI.

1.1 The Epidemiology

Opioid use disorder affects approximately 5.7 million Americans aged 12 and older. In 2023, more than 80,000 Americans died from an opioid-involved overdose. Medication for opioid use disorder is recognized as the standard of care, with three FDA-approved options: methadone, buprenorphine, and naltrexone. The 2020 ASAM National Practice Guideline and the 2020 SAMHSA Treatment Improvement Protocol 63 both endorse MOUD as first-line treatment.[1] Yet only approximately 22 percent of Americans with OUD receive any form of MOUD in a given year, and among those who do, retention is the central clinical challenge.[2]

1.2 The Retention Crisis

Across the published literature, six-month retention on buprenorphine is approximately 50 percent or less. A 2019 review by Shulman, Wai, and Nunes summarized: "Across trials, retention in treatment for buprenorphine is in the range of 50 percent or less at 6 months, highlighting the need for strategies to improve long-term adherence to buprenorphine."[3] A 2019 systematic review by Timko and colleagues found 6-month retention rates ranging from 3 percent to 88 percent across randomized controlled trials, with a median substantially below the threshold required for clinical benefit.[4]

Real-world data are similarly stark. Williams and colleagues, analyzing electronic health records from more than 5,000 buprenorphine-treated OUD patients across 23 substance use programs in 11 US states, reported retention rates of 50 percent at 2 months, 27 percent at 6 months, and 14 percent at 12 months in outpatient settings.[5] The Wakeman et al. comparative effectiveness study published in JAMA Network Open in 2020 reinforced what every MAT clinician already knows: each additional month a patient stays on buprenorphine substantially reduces the probability of overdose and serious opioid-related acute care events.[6]

WHAT THIS MEANS FOR BAYMARK

If a typical BAART clinic has 200 active OUD patients on buprenorphine, the published baseline implies that approximately 100 will discontinue within six months and only about 30 will remain at twelve months. Each discontinuation represents not only a clinical setback but a lost recurring revenue event for the clinic and a substantially elevated overdose risk for the patient. Any intervention that produces even a 10–15 percentage-point absolute improvement in 6-month retention has a transformative effect on both clinical outcomes and clinic economics.

1.3 Why Behavioral Adjuncts to MOUD Matter

The 2020 ASAM National Practice Guideline explicitly states that MOUD is most effective when combined with psychosocial treatment. The Manhapra et al. analysis of more than 33,000 commercially insured buprenorphine patients found that receipt of psychotherapy was associated with significantly increased retention in MAT.[7] The challenge is that psychosocial treatment is exactly what is hardest to deliver consistently in high-volume MAT clinics. Clinician time is finite. Manualized CBT requires training, fidelity monitoring, and supervision. Contingency management requires logistical infrastructure and creates billing complexity. Family interventions require therapist hours that do not exist in most outpatient settings.

This is the operational gap that digital therapeutics are designed to fill. The platform delivers the psychosocial intervention with fidelity. The clinician oversees, intervenes when the data flags risk, and reclaims time previously spent on manual documentation. The patient accesses evidence-based content on their own time, in their own environment, with engagement mechanics that drive sustained use.

1.4 The Specific Operational Reality at BayMark

BayMark operates more than 300 clinics across the United States and Canada, the largest OUD-focused treatment footprint in North America. Its California BAART clinics serve a Medicaid-heavy patient population through Drug Medi-Cal Organized Delivery System (DMC-ODS) contracts and operate under California Department of Health Care Services oversight. DHCS quality reporting increasingly requires structured outcomes data that traditional paper-based clinic workflows cannot easily produce.

This creates a specific opportunity that does not exist in smaller or non-Medicaid OUD providers. BayMark's leverage with DHCS, its ability to negotiate enhanced MAT reimbursement, and its credibility with its private equity owner Webster Capital all depend on the organization's ability to demonstrate measurable outcomes at scale. VictoryAI is designed to generate exactly the data DHCS, payers, and the operating committee want to see — and to do so as a byproduct of patient engagement rather than as an additional clinician documentation burden.

SECTION 02

Evidence-Based Foundations of the VictoryAI Platform

This section reviews the published evidence for each of the six therapeutic mechanisms implemented in VictoryAI. For each mechanism, the section identifies the foundational randomized controlled trials, the most relevant systematic reviews and meta-analyses, the effect sizes reported, the limitations of the evidence base, and the specific way the mechanism is operationalized in the VictoryAI platform.

This is the section BayMark's Chief Medical Officer should read most carefully. Every quantitative claim is cited. Every cited paper is listed in the References section at the back of this dossier with full bibliographic information including DOI or PubMed identifier, so that the citations can be independently verified.

2.1 Digital Cognitive Behavioral Therapy for Substance Use Disorder

Foundational Evidence

Cognitive behavioral therapy is among the most extensively validated psychosocial treatments for substance use disorder. The translation of CBT into a computer-delivered format was pioneered by Kathleen Carroll and colleagues at Yale, who developed Computer-Based Training for Cognitive Behavioral Therapy (CBT4CBT). The original 2008 randomized controlled trial, published in the American Journal of Psychiatry, randomized 77 individuals seeking treatment for substance dependence to standard treatment versus standard treatment plus CBT4CBT. Participants in the CBT4CBT condition submitted significantly more drug-negative urine specimens and showed longer continuous periods of abstinence during treatment.[8] A 6-month follow-up published in 2009 demonstrated durability of effect.[9]

Carroll and colleagues subsequently replicated and extended these findings in cocaine-dependent patients maintained on methadone in a 2014 trial published in the American Journal of Psychiatry, demonstrating that CBT4CBT remained efficacious in patients with OUD already engaged in MOUD.[10] In 2018, Kiluk and colleagues published a three-arm randomized controlled trial comparing treatment as usual, clinician-delivered CBT, and CBT4CBT delivered as a virtual stand-alone with minimal clinical monitoring. The CBT4CBT condition was associated with the best learning of cognitive and behavioral concepts, the highest treatment satisfaction, and continuing benefit over treatment as usual at 6-month follow-up — notably outperforming clinician-delivered CBT on durability.[11]

Most recently, Kiluk and colleagues published a randomized clinical trial of digital CBT for adults with alcohol use disorder in JAMA Network Open in 2024, evaluating the digital CBT program against both clinician-delivered CBT and treatment as usual across an 8-week treatment period with 6-month follow-up. The digital format produced equivalent reductions in percent days drinking compared with clinician-delivered CBT.[12]

Translation to OUD: The reSET-O Evidence Base

The most directly relevant evidence for VictoryAI's design comes from reSET-O, the prescription digital therapeutic developed by Pear Therapeutics. reSET-O delivers a CBT-based intervention derived from the Therapeutic Education System and is indicated as an adjunct to outpatient buprenorphine treatment. In December 2018, the FDA granted reSET-O clearance based on a NIDA-sponsored randomized controlled trial of 170 OUD patients over 12 weeks. The trial demonstrated that patients randomized to reSET-O plus standard buprenorphine treatment had significantly improved retention compared with treatment as usual.[13]

Subsequent real-world evidence has reinforced the original trial findings. A 12-month claims-based analysis published in Advances in Therapy in 2022 by Velez and colleagues compared 901 OUD patients who initiated reSET-O against 978 controls who did not. The analysis found significantly fewer hospital encounters, fewer inpatient stays, and fewer emergency department visits in the reSET-O cohort, with cost reductions of $3,832 per patient over 12 months in the Medicaid subpopulation. Buprenorphine adherence, measured by the medication possession ratio, was significantly increased in the reSET-O cohort.[14] Maricich and colleagues, in a 24-week real-world study published in Hospital Practice in 2021, reported similar improvements in retention and abstinence among reSET-O users.[15]

How VictoryAI Implements Digital CBT

VictoryAI's CBT module, embedded in the patient app as the prescribed Digital Therapeutic for Substance Use, is structured as an 8-module program covering: understanding OUD, MAT as your ally, the opioid trigger map, pain without opioids, rebuilding relationships, understanding relapse, coping skills, and relapse prevention planning. Module content is derived from the evidence-based CBT manuals used in the trials cited above and adapted for the OUD population on MOUD specifically.

Each completed lesson generates a structured event into the provider dashboard. Each completed module generates documentation eligible for billing under H2019 (digital CBT sessions, per 15-minute increment). The clinician sees module completion in real time, can intervene with patients who stall, and can reference specific module content during in-clinic sessions.

Limitations of the Evidence Base

Three limitations of the digital CBT literature are worth noting. First, no published trial of CBT4CBT or reSET-O has lasted longer than 24 weeks (84 days for the reSET-O pivotal trial), so the long-term durability of digital CBT effects beyond six months is not established. Second, real-world evidence for reSET-O comes from claims-based comparison studies rather than long-term randomized trials, and these designs cannot fully control for selection effects. Third, the patient populations in the published trials may not perfectly represent the demographic mix served by BayMark's California BAART clinics, particularly the heavily Medicaid, Spanish-speaking, and unhoused subpopulations. The proposed BayMark pilot is designed to address this third limitation directly.

2.2 Contingency Management for Patients Receiving MOUD

Foundational Evidence

Contingency management (CM) is a behavioral intervention in which patients receive material incentives contingent on objectively verified behavior change — typically, drug-negative urine screens or treatment session attendance. CM is grounded in operant conditioning principles and has been the subject of more than 30 years of randomized controlled trials. Higgins and colleagues at the University of Vermont published the first major CM trials in the early 1990s, and the evidence base has expanded continuously since then.[16]

The single most important piece of evidence for VictoryAI's reinforcement architecture is the 2021 systematic review and meta-analysis by Bolívar and colleagues, published in JAMA Psychiatry. Bolívar et al. identified 74 randomized clinical trials involving 10,444 unique adult participants receiving medication for opioid use disorder, and assessed the effect of contingency management on six clinical outcomes: stimulant use, polysubstance use, illicit opioid use, cigarette smoking, therapy attendance, and medication adherence. The meta-analysis found that contingency management was associated with statistically significant improvements across all six outcome domains. The authors concluded: "These results provide evidence supporting the use of contingency management for addressing common and serious clinical problems among patients receiving medication for opioid use disorder."[17]

A 2021 meta-analysis by Ginley and colleagues, published in the Journal of Consulting and Clinical Psychology, examined long-term outcomes of contingency management based on objective indicators of abstinence. The analysis found that benefits of contingency management persisted well after incentives ended, with participants maintaining outcomes as good as or better than comparison groups for up to one year post-treatment — directly addressing the "will it work after you stop paying them" objection that has historically slowed CM adoption.[18]

Smartphone-Delivered Contingency Management

Two recent studies have specifically examined smartphone-app-based contingency management for OUD. DeFulio and colleagues published a 2024 retrospective cohort study analyzing data from 600 individuals receiving MOUD across opioid treatment programs in Texas. Patients who chose MOUD plus app-based CM (delivered via the WEconnect platform) reported significantly fewer days of opioid use at the end of treatment compared with those who chose MOUD only (mean 8.4 vs. 12.0 days). Retention analysis showed that patients in the MOUD-plus-app-based-CM group were significantly more likely to remain in treatment longer than those in the MOUD-only group.[19]

In 2022, the HHS Office of Inspector General issued Advisory Opinion 22-04, concluding that a digital contingency management program would not violate the federal Anti-Kickback Statute or Beneficiary Inducement Prohibition. This was the first time OIG provided such approval for a

nationally accessible CM program offering full-value monetary rewards, and it removed the single largest regulatory barrier to deploying digital CM in Medicaid populations.[20]

How VictoryAI Implements the Reinforcement Mechanism

VictoryAI does not deliver cash rewards, gift cards, or other direct monetary incentives — and therefore does not invoke the regulatory complexities surrounding traditional contingency management. Instead, VictoryAI implements the same operant-conditioning principle through three integrated reinforcement systems:

- Variable-ratio in-app rewards (XP, level progression, achievement badges) tied to clinically meaningful behaviors: MAT dose confirmations, daily check-ins, COWS assessments, CBT module completion, drug screen attendance, and group session check-ins.
- Family-pledged real-world rewards via the Family Reward Board, where family members commit to specific incentives — meals, experiences, gifts — tied to specific recovery milestones. The reinforcer is delivered by the patient's own family network, not by the provider, which entirely sidesteps the Anti-Kickback Statute concerns that have historically slowed traditional CM adoption.
- Loss-aversion mechanics (the Recovery Flame, Streak Shields, daily quests) that recruit the same operant principles in a way that reinforces continuous engagement rather than discrete behaviors.

Limitations and Honest Caveats

VictoryAI's reinforcement architecture is inspired by contingency management evidence, but it is not contingency management in the strict sense studied in the Bolívar et al. meta-analysis. The published CM trials use monetary or near-monetary incentives delivered by clinical staff for behaviors verified by objective biological measures (typically urine toxicology). VictoryAI uses non-monetary in-app rewards and family-mediated real-world rewards for self-reported and app-verified behaviors. Whether VictoryAI's implementation produces the same effect sizes as the published CM literature is an empirical question that the proposed pilot is designed to answer. The honest interpretation of the evidence is that the underlying behavioral principle is well-supported, the specific implementation is novel, and validation requires the pilot.

2.3 Family-Based Engagement: CRAFT and CRAFT-T

Foundational Evidence

Community Reinforcement and Family Training (CRAFT) was developed by Robert Meyers and colleagues at the University of New Mexico as a structured intervention to help concerned significant others (CSOs) — typically family members — engage treatment-resistant individuals in substance use treatment. CRAFT is grounded in operant principles and emphasizes positive reinforcement of healthy behavior, communication skills, and natural consequences for substance use, while explicitly avoiding the confrontational tactics of older family interventions.[21]

The foundational CRAFT trial, published by Meyers and colleagues in the Journal of Substance Abuse in 1999, evaluated CRAFT with 62 CSOs over 6 months. CSOs completed an average of 87 percent of offered treatment sessions, and 74 percent succeeded in engaging their resistant loved one in treatment.[22] A subsequent randomized comparison of CRAFT against the Johnson Institute Intervention and against Al-Anon/Nar-Anon facilitation, published in the Journal of Consulting and Clinical Psychology in 2002, found CRAFT engagement rates of 64 percent, compared with 30 percent for the Johnson Institute Intervention and just 13 percent for Al-Anon/Nar-Anon facilitation.[23]

A systematic review and meta-analysis of CRAFT trials by Roozen, de Waart, and van der Kroft, published in Addiction in 2010, pooled the available randomized evidence and found that CRAFT engagement rates averaged 67 percent across studies, compared with 18 percent for AA/NA-style approaches and 30 percent for the Johnson Institute Intervention.[24] The CRAFT effect was robust across substance type, CSO-IP relationship type, and study setting.

CRAFT for Opioid Use Disorder Specifically

Brigham and colleagues published a randomized pilot trial of Community Reinforcement and Family Training for Treatment Retention (CRAFT-T) for OUD specifically in Drug and Alcohol Dependence in 2014. The trial enrolled 52 dyads of opioid-dependent adults completing residential buprenorphine detoxification and their CSOs. CRAFT-T showed a significant positive effect on the identified patient's opioid and other drug use, and a moderate-to-large effect on treatment retention, particularly when the CSO was a parent.[25]

Beyond the engagement and retention effects, CRAFT also produces measurable improvements in CSO well-being. Across multiple trials, CSOs receiving CRAFT report reductions in depression, anxiety, anger, and relationship conflict, with improvements in family cohesion.[26] This matters for VictoryAI's design because the family member is not just an instrument for engaging the patient — the family member is also a beneficiary of the intervention, which is critical for sustained family participation over the 90-day pilot and beyond.

How VictoryAI Implements Family-Based Engagement

VictoryAI's Family Reward Board operationalizes the core CRAFT mechanism — positive reinforcement of recovery behavior by family members — through a structured digital interface. Family members are invited by the patient (with explicit consent and 42 CFR Part 2 compliance) and pledge specific real-world rewards tied to specific recovery milestones: a fishing trip at 90 days clean, a family dinner at 30 days, a trip at 365 days. The patient sees the pledged rewards continuously in the app, which functions as a persistent reminder of the family-supported reasons for staying in treatment.

The reciprocity loop is closed by the Gratitude Gifts feature: at milestones, the patient can send personalized digital cards back to the family member who pledged the reward. This is a deliberate implementation of the CRAFT principle that family-based engagement should improve the relationship in both directions, not just the patient's outcomes.

Limitations

CRAFT was developed for face-to-face, manualized delivery by trained clinicians. VictoryAI's Family Reward Board does not replicate the full CRAFT protocol — it does not provide CSO training in communication skills, functional analysis of substance use, or safety planning. It implements the reinforcement-and-engagement component while leaving the broader CSO support architecture to BayMark's existing clinical infrastructure. The hypothesis is that the structured digital reinforcement layer will produce a meaningful subset of the CRAFT benefit; whether that hypothesis holds for the specific BAART California population is an empirical question.

2.4 Remote Therapeutic Monitoring and Digital Engagement

Background

Remote Therapeutic Monitoring (RTM) is a category of CPT codes (98975–98981) introduced by CMS in 2022 to reimburse providers for the collection and management of non-physiological patient data, including data from mobile applications used to monitor adherence to therapy, response to therapy, and the musculoskeletal or respiratory systems. The 2022 expansion explicitly included behavioral health applications, and in 2023 CMS clarified that mental and behavioral health monitoring is an eligible RTM use case.

RTM is a reimbursement framework, not a clinical intervention per se. The clinical intervention is the structured engagement with the digital tool itself. Evidence for the engagement-as-intervention model comes from multiple sources, most notably the Therapeutic Education System trials and the broader literature on digital phenotyping.

Therapeutic Education System Evidence

The Therapeutic Education System (TES) is a web-based behavioral intervention developed by Lisa Marsch and colleagues that combines CBT-based skill-building modules with structured engagement tracking. Campbell and colleagues, in a multisite randomized controlled trial published in the American Journal of Psychiatry in 2014, evaluated TES as an adjunct to standard outpatient SUD treatment across ten community-based programs in the National Drug Abuse Treatment Clinical Trials Network. The trial enrolled 507 patients and found that TES plus reduced-intensity standard treatment produced abstinence rates equivalent to or better than full-intensity standard treatment alone, with TES patients showing significantly improved treatment retention and abstinence outcomes.[27]

Christensen and colleagues, in a 2014 trial published in the Journal of Consulting and Clinical Psychology, specifically evaluated the addition of TES to an efficacious treatment package for opioid dependence. The trial demonstrated that adding the digital therapeutic component improved abstinence outcomes beyond what was achieved by an already-effective in-person treatment package.[28]

Engagement-Outcome Relationships

The mechanism by which structured digital engagement improves outcomes is, broadly, that consistent contact with treatment-relevant content and clinical monitoring keeps patients connected to recovery during the periods between in-clinic visits — which is when most relapses occur. Across the digital therapeutics literature, dose-response relationships are consistently observed: patients who engage more with the digital intervention show better outcomes than those who engage less. This relationship is the basis for the RTM billing threshold of 16 or more days of monitoring per month.

How VictoryAI Implements RTM

VictoryAI is designed to clear the 16-day RTM threshold as a routine outcome of normal patient engagement. Daily MAT confirmation, daily check-ins, weekly CBT module work, and the gamification mechanics that drive return visits combine to produce an average engagement profile that typically exceeds 20 active days per month. The patient app is the data-capture layer; the provider dashboard is the management layer; the EMR export is the documentation layer that allows the clinic to claim 98977 (RTM device supply) and 98980/98981 (RTM treatment management) reimbursement on top of their existing buprenorphine billing.

Limitations

RTM is a young billing category, and the operational guidance from CMS continues to evolve. VictoryAI's claim that engagement events generate RTM-eligible documentation is based on the current 2024–2025 CMS coding guidance and the published interpretations from major billing consultancies. BayMark's billing team should verify the specific code applicability against its own payer mix during pilot setup. The clinical evidence that engagement improves outcomes is strong; the operational evidence that any specific digital tool produces consistent, audit-ready RTM documentation requires per-deployment validation, which the pilot is designed to provide.

2.5 Voice Biomarkers and Digital Phenotyping

The Underlying Science

Voice carries detectable acoustic and linguistic markers of psychological state. Depression, anxiety, and mood deterioration produce measurable changes in pitch, prosody, speech rate, pause duration, and lexical complexity. The detection of these markers from short audio samples using machine learning has been the subject of growing scientific interest over the past decade. A 2025 systematic review published in the Journal of Voice analyzed 12 studies examining 16,872 participants and found that AI-driven speech and voice quality outcomes consistently distinguished depression from healthy controls with area under the curve values ranging from 0.71 to 0.93.[29]

A 2024 study published in BMC Psychiatry examined automated speech analysis in 96 participants (44 with major depressive disorder and 52 healthy controls), identifying discriminating acoustic features including speech rate, pitch variability, and prosodic measures that correlated significantly with both the Beck Depression Inventory and the Hamilton Rating Scale for Depression.[30]

Smartphone-Based Voice Monitoring

The translation of voice biomarker science to smartphone-based deployment has accelerated since 2022. Larsen and colleagues, in a 2024 study published in Frontiers in Psychiatry, evaluated a smartphone app that collected 30-second voice samples and analyzed them for vocal biomarkers including jitter, shimmer, pitch variability, energy variability, vowel space, phonation duration, speech rate, and pause duration. The study found that the app's mental fitness vocal biomarker scores were significantly associated with depression symptom severity, with relative risk ratios of 2.6 for time-weighted analyses.[31]

A 2025 study published in NPJ Digital Medicine analyzed 3,151 weekly voice diaries from 284 German-speaking adults (128 with MDD and 156 controls) and demonstrated that sentence-embedding-based language models extracted from voice diaries could predict Beck Depression Inventory scores with mean absolute errors as low as 4.37 points. The authors specifically noted that voice diaries collected in ecologically valid contexts "offer a scalable, non-invasive lens on mood" that addresses the limitation of infrequent clinic-based self-report assessment.[32]

Why Voice Matters for OUD Specifically

Mood deterioration is one of the strongest predictors of relapse in OUD recovery. Patients in the days leading up to a relapse event typically show measurable declines in mood, sleep quality, and engagement with positive activities. Self-report screeners (PHQ-9, GAD-7) administered monthly or quarterly cannot capture these short-term fluctuations; daily self-reported mood is subject to reporting bias and selective non-completion. Voice diaries, recorded as a routine

engagement behavior in the patient app, provide a passive and continuous signal that can flag pre-relapse trajectories before the patient is consciously aware of the trajectory themselves.

How VictoryAI Implements Voice Biomarkers

The Voice Journal feature in the patient app invites the patient to record short, free-form audio entries describing their day, their mood, or anything they wish to reflect on. Audio is processed locally on the device for natural language sentiment analysis and acoustic feature extraction; raw audio is never transmitted to VictoryAI servers and is deleted from the device after feature extraction. The extracted features — sentiment score, prosodic markers, lexical complexity — are transmitted to the provider dashboard as a structured event, where they contribute to the AI predictive layer described in Section 3.

This privacy-preserving architecture is critical. Voice is identifiable in a way that text is not, and 42 CFR Part 2 imposes strict limits on the handling of substance use disorder patient data. By processing voice on-device and transmitting only features (not raw audio), VictoryAI captures the predictive value of the signal without creating the privacy and regulatory risk that would come with transmitting and storing patient voice recordings.

Limitations

Voice biomarker science is still in active development. The reviews cited above note methodological heterogeneity, dataset bias, and limited longitudinal evidence as ongoing challenges. The 2022 Lancet editorial on vocal biomarkers explicitly stated that better ethical and technical standards are required before AI-powered vocal biomarkers can be endorsed for routine clinical use. VictoryAI treats voice biomarkers as one signal among many in the AI predictive layer — not as a stand-alone diagnostic — which is the appropriate level of evidentiary caution given the current state of the field. The pilot will produce population-specific data on whether voice features add predictive value over MAT adherence and engagement signals alone.

2.6 MAT Adherence and the Retention Crisis

The Evidence on Adherence Interventions

Improving MAT adherence is the central clinical challenge in OUD treatment. The Manhapra et al. analysis of more than 33,000 commercially insured buprenorphine patients found that one-year retention on buprenorphine averaged just 28 percent, with substantial variation by patient characteristics, and that receipt of psychotherapy was associated with significantly increased retention.[7]

Interventions specifically designed to improve MOUD adherence have been the subject of multiple recent trials. Wakeman and colleagues, in a comparative effectiveness study of OUD treatment pathways published in JAMA Network Open in 2020, analyzed national insurance claims data and found that buprenorphine and methadone were the only treatment pathways associated with reductions in overdose and serious opioid-related acute care, and that the protective effect grew substantially with each additional month of medication continuity.[6]

Why Daily Confirmation Matters

Buprenorphine missed doses are not benign. Even brief lapses in dosing can produce mild withdrawal, intensified cravings, and increased relapse risk. Patients who miss multiple consecutive doses are at substantially elevated risk of leaving treatment entirely. Yet most outpatient buprenorphine programs have no real-time visibility into between-visit dosing — clinicians learn about missed doses only at the next scheduled visit, which may be a week or more later.

VictoryAI's MAT Tracker is designed to address this visibility gap. Patients confirm each dose with a single tap, earning XP and contributing to their Recovery Flame streak. Missed doses generate a structured event into the provider dashboard within hours, not weeks. Clinicians can intervene with a phone call, a Sage AI nudge, or an in-clinic check before a missed dose becomes a missed week becomes a treatment dropout.

How the Engagement System Reinforces Adherence

The MAT Tracker is the most heavily reinforced action in the entire VictoryAI engagement architecture. Each confirmed dose earns 50 XP — the highest single-action XP value in the platform — and contributes to multiple overlapping reinforcement systems: the daily streak counter, the weekly adherence calendar, the Recovery Flame, the family-pledged reward thresholds, and the level progression system. This is a deliberate design choice. The single behavior with the largest impact on patient outcomes also has the largest reinforcement weight.

Limitations

Self-reported MAT confirmation is not the same as biologically verified dosing. A patient can tap the confirmation button without actually taking the dose. VictoryAI's mitigation is not technical (we do not impose biometric verification on already-burdened patients) but clinical: the engagement system is designed to make confirmed-but-not-actually-taken dosing a self-defeating strategy, because the same patient app captures craving levels, urge mode activations, voice journal sentiment, and COWS scores — all of which would deteriorate if the patient were not actually taking buprenorphine. The provider dashboard surfaces inconsistencies between MAT confirmations and the surrounding clinical signals so that clinicians can investigate. The pilot will generate data on the rate of dosing discrepancies and the dashboard's ability to detect them.

SECTION 03

How VictoryAI Implements Each Evidence Base

This section consolidates the mechanism-to-feature mapping for ease of reference. For each of the six evidence-based mechanisms reviewed in Section 2, the table below identifies the specific VictoryAI features that operationalize the mechanism, the primary published evidence that supports it, and the dashboard view where the resulting clinical data surfaces for BayMark clinicians.

Mechanism	VictoryAI Feature(s)	Key Evidence	Dashboard View
Digital CBT	8-module CBT4SUD program; CBT Cliffhangers; lesson tracking	Carroll 2008, 2014; Kiluk 2018, 2024; reSET-O FDA pivotal trial 2018	Workbook / DTx Progress
Reinforcement (CM-derived)	XP system; Recovery Flame; Family Reward Board; Mystery XP; Daily Quests	Bolívar 2021 JAMA Psychiatry meta-analysis; Ginley 2021; DeFulio 2024	Engagement & Gamification
Family-Based (CRAFT)	Family Reward Board; Gratitude Gifts; family member onboarding	Meyers 1999; Brigham 2014 CRAFT-T; Roozen 2010 systematic review	Engagement (top driver subview)
RTM & Engagement	Daily check-ins; MAT Tracker; full event taxonomy	Campbell 2014 (TES); Christensen 2014; CMS 2022 RTM rules	EMR Data Export
Voice Biomarkers	Voice Journal with on-device NLP; sentiment trajectory	Larsen 2024; multiple 2024–2025 systematic reviews	Mood & Sentiment, AI Predictive
MAT Adherence	MAT Tracker; missed-dose alerts; clinical alert escalation	Manhapra 2018; Wakeman 2020; Shulman 2019 review	MAT Adherence Analytics

The Integration Hypothesis

The published evidence base supports each mechanism individually. No published trial has tested the integration of all six mechanisms in a single digital platform purpose-built for high-volume MAT clinics. This is the empirical gap VictoryAI is designed to fill, and it is the gap the proposed BayMark pilot is designed to test.

The hypothesis is that integration produces additive — and possibly multiplicative — benefits over any single mechanism deployed alone. The reasoning is that each mechanism addresses a different failure mode in current MAT delivery: CBT addresses skill deficits, contingency-

management-derived reinforcement addresses motivation, family engagement addresses social context, RTM addresses between-visit invisibility, voice biomarkers address late detection of mood deterioration, and MAT adherence support addresses the central clinical event itself. A platform that addresses all six failure modes simultaneously should outperform a platform that addresses any one of them.

Whether that hypothesis is correct, and by how much, is exactly what the pilot is designed to determine. The integration hypothesis is plausible; it is not yet proven. This dossier asks BayMark to find the hypothesis plausible enough to justify the empirical test — not to accept it as established.

SECTION 04

The Comparative Landscape

BayMark is being approached by multiple digital health vendors. The honest framing of this dossier requires acknowledging where VictoryAI sits relative to the competitive landscape. This section identifies the most relevant comparators and describes how VictoryAI positions itself.

Pear Therapeutics (reSET-O)

reSET-O is the most directly comparable platform and is the primary comparator throughout this dossier. It is the only FDA-cleared prescription digital therapeutic for OUD specifically, and its evidence base is the strongest of any digital tool in the category. Pear Therapeutics filed for Chapter 11 bankruptcy in 2023, and the future of reSET-O distribution is uncertain. The clinical evidence base for reSET-O remains valid regardless of the corporate situation.

VictoryAI's position relative to reSET-O: VictoryAI is broader in scope (six mechanisms vs. two), is integrated with EMR billing workflows in a way reSET-O was not, includes the family-based engagement layer that reSET-O lacks, and is designed for the operational reality of high-volume MAT providers like BayMark rather than as a prescription-only tool requiring individual physician prescription. VictoryAI is not currently FDA-cleared. The pilot would generate the data required for a future FDA submission if BayMark and VictoryAI determined that path was strategically worthwhile.

CHESS Health (A-CHESS, Connections, Recovery Coach)

CHESS Health offers a suite of recovery support apps and a contingency management platform (Rewards Engine). The A-CHESS app has been studied in randomized controlled trials for alcohol use disorder, and the broader CHESS suite is deployed in multiple state recovery initiatives. CHESS does not have an OUD-specific FDA clearance. VictoryAI's positioning relative to CHESS is the same OUD-specific focus and the same EMR-integrated billing workflow that distinguish it from reSET-O.

Bicycle Health

Bicycle Health is a virtual buprenorphine clinic — not a digital adjunct to in-person treatment, but a full telehealth treatment provider. It is a competitor to BayMark, not a vendor to BayMark. Bicycle Health's existence is itself one of the reasons VictoryAI represents a strategic opportunity for BayMark: a tech-enabled OUD competitor is taking patients out of the in-person market, and BayMark's defense against that erosion is to become the most data-rich, outcomes-transparent in-person provider in the country. VictoryAI is the operational mechanism for that defense.

Workit Health, Ophelia, Boulder Care

Like Bicycle Health, these are full-service virtual OUD treatment providers, not vendor platforms. Each is taking market share that would otherwise flow to traditional in-person MAT providers. Each uses some form of mobile patient engagement, but none publishes the structured engagement-to-billing-code mapping that VictoryAI is built around.

Big Health (Sleepio, Daylight)

Big Health's Sleepio for insomnia and Daylight for anxiety are not OUD-specific tools, but they are the most successful examples of the digital therapeutics commercialization playbook that VictoryAI is following. Big Health secured early traction by submitting Sleepio to NICE in the United Kingdom and to ICER in the United States, generating health-economic evidence that supported payer adoption. Sleepio is now reimbursed by multiple US health plans and integrated into multiple large employer benefit programs. Big Health's path from clinical evidence to payer adoption is the template VictoryAI is following — adapted for the OUD context and for the specific operational reality of MAT providers.

The Honest Comparative Summary

WHERE VICTORYAI SITS TODAY

VictoryAI has the broadest mechanism integration of any platform in the OUD digital health category, the most operationally relevant EMR-billing-code mapping, and the only purpose-built provider analytics dashboard designed for the workflow of high-volume MAT clinics. VictoryAI does not yet have its own clinical evidence base, an FDA clearance, or a published outcomes study. The proposed BayMark pilot is the path by which VictoryAI moves from "strongest mechanism integration" to "strongest evidence-backed mechanism integration."

SECTION 05

The Proposed Clinical Study Protocol

This section presents the IRB-ready study protocol that would govern the BayMark joint innovation pilot. The protocol is structured to meet the methodological standards of peer-reviewed journals in the addiction medicine field, including the American Journal of Psychiatry, JAMA Psychiatry, JAMA Network Open, Drug and Alcohol Dependence, and the Journal of Substance Use and Addiction Treatment.

5.1 Study Design

The pilot is designed as a prospective, single-arm, open-label feasibility and effectiveness study with a contemporaneous matched comparison group drawn from BayMark's standard-of-care patient population at non-pilot BAART California sites. This design is selected over a randomized controlled trial for the pilot phase because (a) it is appropriate to the feasibility-and-acceptability stage of evidence development, (b) it minimizes patient burden during the initial deployment, and (c) it produces effect-size estimates that can power a subsequent randomized trial if the pilot results justify one.

The pilot is not a stand-alone definitive trial. It is the first stage of a planned two-stage development program: (1) a 90-day feasibility-and-effectiveness pilot at three to five California BAART sites with two hundred or more patients, generating the engagement, adherence, and outcomes data needed to inform a Stage 2 study, and (2) a planned subsequent randomized controlled trial of expanded scope, conditional on pilot results.

5.2 Hypotheses

Primary Hypothesis

Patients enrolled in VictoryAI plus standard BayMark MAT will demonstrate higher 90-day buprenorphine adherence (measured by medication possession ratio) than the matched comparison cohort receiving standard BayMark MAT alone.

Secondary Hypotheses

5. Patients enrolled in VictoryAI will demonstrate higher 90-day treatment retention than the matched comparison cohort.
6. Patients enrolled in VictoryAI with active Family Reward Board engagement will show higher retention than VictoryAI patients without family engagement.
7. Voice journal sentiment trajectory will be associated with subsequent self-reported relapse events at 5–7 day lead time.

8. Each pilot patient will generate at least $300 per month in EMR-eligible billable data across the RTM, DTx, and OUD treatment code categories.
9. At least 80 percent of credentialed BayMark clinicians at pilot sites will use the provider dashboard at least once per week throughout the pilot.

5.3 Inclusion and Exclusion Criteria

Inclusion

- Adults aged 18 years or older
- Active enrollment at a participating BAART California clinic for medication for opioid use disorder (buprenorphine, methadone, or naltrexone)
- Diagnosis of opioid use disorder per DSM-5-TR criteria
- Access to a personal smartphone (iOS or Android) with cellular data or reliable WiFi
- Ability to provide informed consent in English or Spanish
- Willingness to use the VictoryAI patient app for the duration of the 90-day pilot

Exclusion

- Current acute psychiatric crisis requiring inpatient admission
- Active suicidal ideation with intent or plan at baseline screening
- Cognitive impairment that would prevent informed consent or app use
- Anticipated relocation outside the BAART service area within 90 days
- Participation in another digital health clinical trial within the past 30 days

5.4 Outcome Measures

Primary Outcome

Buprenorphine medication possession ratio at 90 days, computed from BayMark prescription fill data and VictoryAI MAT confirmation events. Analysis will be presented as both the conventional MPR (refills divided by days) and the engagement-augmented MPR (which includes self-reported daily confirmations as a secondary signal).

Secondary Outcomes

- 90-day treatment retention (binary: still enrolled at day 90)
- Average daily craving score (0–10 scale) over the 90-day period
- Number of urge mode activations and resolution path
- CBT module completion rate
- PHQ-9 and GAD-7 trajectory from baseline through day 90
- COWS score trajectory for patients with documented withdrawal events
- Number of family members connected per patient
- Family-pledged rewards claimed during the pilot

- Self-reported relapse events (with the 5–7 day voice biomarker prediction window evaluated post hoc)
- Total RTM-eligible engagement days per patient per month
- Total documented billable code data generated per patient per month

Process Outcomes

- Patient app session count, duration, and feature usage breakdown
- Provider dashboard usage by clinician role
- Time from clinical alert generation to clinician action
- Technical reliability (uptime, sync latency, support tickets)

5.5 Sample Size and Statistical Analysis

The pilot is designed to enroll a minimum of 200 patients across 3 to 5 BAART California sites, with a target enrollment of 250 to allow for attrition. With a baseline 6-month buprenorphine retention rate of approximately 50 percent (per Shulman 2019 and the Williams 2019 real-world data) and a hypothesized improvement to 65 percent, a sample of 200 provides 80 percent power to detect a 15 percentage-point absolute difference in retention at alpha 0.05 (two-tailed) compared with the matched control cohort, assuming a control cohort of equal size.

The matched comparison cohort will be drawn from non-pilot BAART California sites using propensity score matching on age, sex, race/ethnicity, MOUD type, baseline MOUD duration, primary substance, and Charlson comorbidity index. Matching will be performed in coordination with BayMark's analytics team using BayMark's existing electronic health record data.

Primary analysis will be intention-to-treat. A per-protocol analysis will be conducted as a secondary analysis, defining per-protocol as completion of the VictoryAI onboarding plus at least 16 active engagement days per month. Statistical analyses will be performed in R (version 4.4 or later) with code and analysis scripts shared between the VictoryAI clinical team and BayMark's analytics team.

5.6 Safety Monitoring

Patient safety during the pilot is governed by a dedicated safety protocol with the following elements. First, the Voice Journal NLP module includes a safety classifier that flags any entry containing language indicative of suicidal ideation, self-harm intent, intent to relapse with high-risk substances, or domestic violence. Flagged entries trigger an immediate alert to the patient's assigned BayMark clinician within 15 minutes. Second, the in-app Crisis / SOS button is persistent on every screen and connects directly to 911, the 988 Suicide and Crisis Lifeline, the SAMHSA National Helpline, and the Crisis Text Line. Third, a dedicated VictoryAI clinical safety officer (a licensed clinician) is available during US business hours throughout the pilot to coordinate with BayMark on any escalations. Fourth, the pilot will be monitored by a Data and Safety Monitoring

Board (DSMB) composed of representatives from VictoryAI, BayMark, and at least one independent clinical reviewer.

Adverse events will be classified per the standard categories used in addiction medicine clinical research and reported to the DSMB on a monthly basis throughout the pilot. Serious adverse events (overdose, suicide attempt, hospitalization, or death) will be reported to the DSMB within 24 hours of identification. The DSMB has authority to halt the pilot if safety concerns warrant.

SECTION 06

Risk Mitigation and Patient Safety

BayMark's clinical leadership will rightly evaluate this pilot through a risk lens. This section identifies the specific risks the pilot creates for patients and for BayMark, and the specific mitigations VictoryAI commits to.

6.1 Patient-Level Risks

Risk: A patient experiences an acute crisis and the app fails to escalate

Mitigation: The Voice Journal NLP safety classifier, the persistent SOS button, the auto-escalation of high COWS scores and high craving events, and the manual escalation path through Sage AI all provide redundant escalation channels. The clinical safety officer monitors flagged events during business hours; outside business hours, the SOS button connects directly to 988, 911, and the Crisis Text Line, none of which depend on VictoryAI infrastructure.

Risk: A patient becomes overly reliant on the app and disengages from in-person clinical care

Mitigation: VictoryAI is designed as an adjunct, not a replacement. Every screen in the patient app reinforces in-clinic appointments. The Appointments feature in the app integrates with BayMark's scheduling system to remind patients of upcoming visits. The provider dashboard surfaces patients whose app engagement is high but whose in-clinic attendance is low, so that BayMark clinicians can intervene before the disengagement becomes problematic.

Risk: A patient's family member abuses the Family Reward Board

Mitigation: Family member onboarding requires explicit patient consent, and the patient can revoke any family connection at any time without explanation. Family members do not see clinical data — they see only the milestones the patient has achieved and the rewards they themselves have pledged. The patient sees all family interactions and can flag any family member as inappropriate, which immediately removes that member from the patient's network and notifies the patient's clinician.

Risk: 42 CFR Part 2 confidentiality is breached

Mitigation: VictoryAI's compliance architecture implements 42 CFR Part 2 segmentation at the data layer, not just the access layer. SUD-related data is tagged at point of capture and routed through a separate consent verification flow before any disclosure. The Data Processing Agreement and Business Associate Agreement governing the pilot will specify the exact data

flows and the consent requirements for each. A compliance audit will be conducted prior to pilot launch and again at pilot midpoint.

6.2 BayMark Organizational Risks

Risk: The pilot generates negative results that damage BayMark's reputation

Mitigation: The publication and dissemination plan in Section 7 commits to publishing pilot results regardless of outcome. Negative or null results, properly contextualized, do not damage BayMark's reputation — they enhance it by demonstrating BayMark's commitment to evidence-based practice. The risk of NOT publishing negative results is greater than the risk of publishing them, both for scientific integrity and for BayMark's standing with DHCS and academic partners.

Risk: Pilot operations disrupt BAART clinic workflow

Mitigation: VictoryAI provides dedicated implementation support throughout the pilot, with a named customer success lead assigned to each pilot clinic. Clinician training is delivered before pilot launch, with refresher sessions at weeks 2, 4, 8, and 12. Daily check-ins with the clinical lead at each pilot site are conducted during the first two weeks, transitioning to weekly check-ins for the remainder of the pilot.

Risk: Technical failures interrupt patient care

Mitigation: VictoryAI commits to 99.5 percent uptime during the pilot, with documented incident response procedures. Patient-facing app failures fall back to a local-only mode that preserves all data for resync. Provider dashboard failures do not impact patient care because the dashboard is read-only — clinicians retain full access to BayMark's existing Kipu EMR throughout. Any sustained outage longer than 4 hours triggers an immediate notification to all credentialed BayMark users and a same-day post-incident report.

Risk: BayMark invests pilot effort and the platform fails to meet success criteria

Mitigation: The pilot agreement includes the enterprise credit clause described in the BayMark Pilot Playbook: if BayMark proceeds to enterprise deployment within 180 days of pilot completion, the entire pilot fee is credited against the enterprise license. If the pilot does not meet success criteria, BayMark retains all data generated during the pilot, retains the right to publish the outcomes, and gains the operational learning of having tested an integrated digital health platform at scale — which has standalone value for future vendor evaluations.

SECTION 07

Publication and Dissemination Plan

The clinical evidence dossier for any digital health platform is only as credible as its commitment to peer-reviewed publication of results. This section commits to specific publication and dissemination steps that are independent of pilot outcome — favorable, unfavorable, or null results will all be published.

7.1 Pre-Registration

The pilot study will be pre-registered on ClinicalTrials.gov before the first patient is enrolled. The pre-registration will include the primary hypothesis, secondary hypotheses, sample size, inclusion and exclusion criteria, planned outcome measures, planned statistical analysis approach, and the publication commitment. Pre-registration locks the analytical plan in advance of seeing the data, which protects against post-hoc reanalysis to chase favorable findings.

7.2 Primary Publication

The primary publication will report the 90-day pilot outcomes against the pre-registered primary and secondary hypotheses. The target journals, in order of preference, are: Drug and Alcohol Dependence, Journal of Substance Use and Addiction Treatment, JAMA Network Open, and Addiction. The primary publication will be co-authored by VictoryAI clinical leadership and BayMark clinical leadership, with authorship order to be determined per ICMJE criteria based on contribution. The publication will follow CONSORT-style reporting standards adapted for non-randomized digital health pilots, including the CONSORT-EHEALTH extension where applicable.

7.3 Conference Presentation

Pilot results will be submitted as an abstract for presentation at one or more of the following conferences: the American Society of Addiction Medicine (ASAM) annual meeting, the American Academy of Addiction Psychiatry (AAAP) annual meeting, the College on Problems of Drug Dependence (CPDD), and the Society for Behavioral Medicine. Conference presentation typically occurs 6–9 months after pilot completion, providing the first public visibility for the results in advance of the journal publication timeline.

7.4 Health Economic Reporting

A separate health economic analysis will be conducted using the EMR billing data captured during the pilot, comparing the pre-VictoryAI and during-VictoryAI billing trajectories at participating BAART sites. The health economic analysis will be submitted as a separate publication, with

target journals including Advances in Therapy, Journal of Market Access and Health Policy, and Value in Health. This is the publication that will most directly support BayMark's enterprise rollout business case and any future DHCS or payer negotiations.

7.5 DHCS and Payer Reporting

Pilot data will be made available to BayMark in a format suitable for DHCS quality reporting and for payer negotiations. The format will follow DHCS DMC-ODS reporting templates and will be reviewed for completeness and quality by the joint VictoryAI-BayMark analytics team prior to any external use. BayMark retains full control over what is shared with DHCS or payers — VictoryAI provides the data and the format, but does not communicate directly with regulatory or payer entities without BayMark's explicit approval.

7.6 Negative Results Commitment

AN EXPLICIT COMMITMENT

If the pilot fails to meet its pre-registered primary or secondary endpoints, VictoryAI commits to publishing those negative results within 12 months of pilot completion. The publication commitment is unconditional and is written into the pilot agreement. Selective publication of favorable results is the single most serious form of evidence corruption in clinical research, and VictoryAI's clinical and scientific affairs leadership will not engage in it.

SECTION 08

Limitations and Open Questions

Every clinical evidence dossier should end with an honest accounting of what the evidence does not show and what questions remain open. This section is that accounting.

8.1 What This Dossier Cannot Establish

This dossier cannot establish that VictoryAI improves outcomes for patients with OUD. It cannot establish that the integration of six evidence-based mechanisms produces additive benefits over any single mechanism. It cannot establish that the engagement effects observed in the published literature for individual mechanisms (CBT4CBT, reSET-O, contingency management, CRAFT, voice biomarkers) will be reproduced in the BayMark BAART California population. It cannot establish that the specific implementation choices VictoryAI has made — XP values, family reward mechanics, Recovery Flame design, voice processing architecture — produce the predicted clinical effects.

These are exactly the questions the proposed pilot is designed to answer. The pilot is the bridge between the strong mechanism-level evidence reviewed in Section 2 and the platform-level evidence VictoryAI requires to support its eventual commercial scaling.

8.2 Open Questions the Pilot Will Address

10. Does VictoryAI engagement reach the published RTM threshold (16+ active days/month) in a real-world Medicaid OUD population, or does engagement degrade below the threshold after the first month?
11. Does the Family Reward Board produce meaningful family engagement at scale, or do most patients fail to invite a family member?
12. Does voice biomarker analysis add predictive value over MAT adherence and engagement signals alone for relapse prediction?
13. Are the CBT modules completed at rates comparable to the published CBT4CBT and reSET-O literature, or does the broader feature set distract patients from the core therapeutic content?
14. Do BayMark clinicians find the provider dashboard usable enough to integrate into their daily workflow, or does it become shelfware after the first month?
15. Does the EMR-billable data capture clear the $300/patient/month threshold in real billing practice, or is the gap between captured-data and billable-data larger than projected?
16. Are there demographic subgroups (Spanish-speaking patients, unhoused patients, patients without smartphones, older patients) for whom VictoryAI underperforms or fails entirely?

8.3 Open Questions the Pilot Will Not Address

The pilot is 90 days. It cannot answer questions about long-term durability of effects beyond 90 days. It cannot answer questions about cost-effectiveness over multi-year horizons. It cannot answer questions about whether VictoryAI works in non-MAT settings (residential treatment, intensive outpatient, recovery housing). It cannot answer questions about whether the platform produces sustained improvements after a patient stops using it. These are questions for Stage 2 and Stage 3 development, conditional on pilot results that justify continued investment.

8.4 What an Honest Reader Should Take Away

An honest reader of this dossier should conclude the following. The therapeutic mechanisms VictoryAI implements are individually well-supported by published evidence. The integration of those mechanisms into a single platform is novel and unproven. The proposed pilot is methodologically sound, ethically appropriate, and capable of generating publication-quality evidence on whether the integration produces the predicted clinical effects. The pilot is the right next step. The decision is whether BayMark wants to be the partner that runs that pilot — and the dossier exists to support a reasoned answer to that question, not to manufacture a particular conclusion.

SECTION 09

From Pilot to Series A: The DTx Precedent

This section is included for the benefit of BayMark's clinical leadership in understanding why the joint innovation pilot model is the standard pathway in the digital therapeutics commercialization landscape — and why the clinical evidence dossier exists in the first place.

9.1 The Pattern

Across the digital therapeutics industry, the path from early-stage company to commercial scale follows a remarkably consistent pattern. The company secures one named-brand pilot partner, generates the first wave of platform-specific clinical evidence, publishes that evidence in peer-reviewed journals, uses the published evidence to raise a Series A or Series B at substantially better terms than would have been available without it, and then scales to additional health system customers using the original pilot partner as the lighthouse reference.

This is the pattern Pear Therapeutics followed with reSET and reSET-O, with NIDA-sponsored trials providing the early evidence base and the FDA submission providing the commercial credentialing event. It is the pattern Big Health followed with Sleepio and Daylight, with the NICE submission and the published RCTs providing the credentialing events. It is the pattern Akili followed with EndeavorRx for pediatric ADHD, with the FDA De Novo authorization providing the commercial inflection point. It is the pattern Click Therapeutics, Pear, and Big Health all used to anchor their commercial strategies in a peer-reviewed clinical evidence base before scaling.

9.2 Why BayMark Specifically

BayMark is uniquely positioned to be the lighthouse partner for VictoryAI for four reasons. First, scale: BayMark operates more than 300 clinics, and a successful pilot at 3 to 5 California sites implies a 60-fold national rollout opportunity that no smaller OUD provider can match. Second, regulatory leverage: BayMark's existing relationship with California DHCS and the DMC-ODS framework means that pilot results have a direct path into reimbursement and quality reporting decisions. Third, EMR standardization: BayMark's use of Kipu EMR across its network means that the integration work done during the pilot is directly reusable across the enterprise. Fourth, competitive context: tech-enabled OUD providers like Bicycle Health are taking patients out of the in-person market, and BayMark's defense against that erosion requires a partnership with the most outcomes-rich digital adjunct platform in the category.

9.3 What the Pilot Becomes

If the pilot succeeds and VictoryAI publishes positive results, three things happen for BayMark. First, the published outcomes become the foundation of BayMark's quality story for DHCS and for any future payer negotiations. Second, the enterprise rollout business case becomes self-evident — the pilot fee is credited against the enterprise license, and the per-clinic cost of expansion drops substantially as the platform scales across the BAART California footprint and then to the broader 300-clinic network. Third, BayMark becomes the named co-developer on every subsequent VictoryAI clinical publication, every conference presentation, and every payer case study, which has reputational value that compounds over time.

If the pilot does not succeed, BayMark has still generated the first rigorous outcomes data on an integrated digital health platform in OUD treatment, the data is publishable, and BayMark retains the operational learning of having tested the model. The asymmetry of the pilot's outcomes — significant upside if successful, limited downside if not — is exactly the asymmetry that has historically made the pilot-to-Series-A model so attractive to early DTx companies and their first health system partners.

VictoryAI is asking BayMark to be the lighthouse partner. This dossier is the clinical evidence foundation on which that partnership is being proposed. The science is strong enough to justify the test. The decision is whether BayMark wants to run it.

REFERENCES

References

All citations in this dossier are listed below in the order they appear in the text. Each entry includes sufficient bibliographic information for independent verification, including DOI or PubMed identifier where available.

1. American Society of Addiction Medicine. The ASAM National Practice Guideline for the Treatment of Opioid Use Disorder: 2020 Focused Update. Rockville, MD: ASAM; 2020.
2. Substance Abuse and Mental Health Services Administration. Medications for Opioid Use Disorder. Treatment Improvement Protocol (TIP) Series 63. Rockville, MD: SAMHSA; 2020.
3. Shulman M, Wai JM, Nunes EV. Buprenorphine Treatment for Opioid Use Disorder: An Overview. CNS Drugs. 2019;33(6):567–580. doi:10.1007/s40263-019-00637-z. PMC6585403.
4. Timko C, Schultz NR, Cucciare MA, Vittorio L, Garrison-Diehn C. Retention in Medication-Assisted Treatment for Opiate Dependence: A Systematic Review. J Addict Dis. 2016;35(1):22–35. doi:10.1080/10550887.2016.1100960.
5. Williams AR, Samples H, Crystal S, Olfson M. Acute Care, Prescription Opioid Use, and Overdose Following Discontinuation of Long-Term Buprenorphine Treatment for Opioid Use Disorder. Am J Psychiatry. 2020;177(2):117–124. (See also: Factors That Affect Patient Attrition in Buprenorphine Treatment for OUD: A Retrospective Real-World Study Using Electronic Health Records, PMC8560173, 2021.)
6. Wakeman SE, Larochelle MR, Ameli O, et al. Comparative Effectiveness of Different Treatment Pathways for Opioid Use Disorder. JAMA Netw Open. 2020;3(2):e1920622. doi:10.1001/jamanetworkopen.2019.20622.
7. Manhapra A, Quinones L, Rosenheck R. Three-Year Retention in Buprenorphine Treatment for Opioid Use Disorder Nationally in the Veterans Health Administration. Psychiatric Services. 2018; (See also: Manhapra et al., association of psychotherapy receipt with buprenorphine retention in commercially insured patients, 2018.)
8. Carroll KM, Ball SA, Martino S, et al. Computer-Assisted Delivery of Cognitive-Behavioral Therapy for Addiction: A Randomized Trial of CBT4CBT. Am J Psychiatry. 2008;165(7):881–888. PMC2562873.
9. Carroll KM, Ball SA, Martino S, et al. Enduring Effects of a Computer-Assisted Training Program for Cognitive Behavioral Therapy: A 6-Month Follow-Up of CBT4CBT. Drug Alcohol Depend. 2009;100(1–2):178–181.
10. Carroll KM, Kiluk BD, Nich C, et al. Computer-Assisted Delivery of Cognitive-Behavioral Therapy: Efficacy and Durability of CBT4CBT Among Cocaine-Dependent Individuals Maintained on Methadone. Am J Psychiatry. 2014;171(4):436–444.
11. Kiluk BD, Nich C, Buck MB, et al. Randomized Clinical Trial of Computerized and Clinician-Delivered CBT in Comparison With Standard Outpatient Treatment for Substance Use Disorders: Primary Within-Treatment and Follow-Up Outcomes. Am J Psychiatry. 2018;175(9):853–863. doi:10.1176/appi.ajp.2018.17090978. PMC6120780.
12. Kiluk BD, DeVito EE, Buck MB, et al. A Digital Cognitive Behavioral Therapy Program for Adults With Alcohol Use Disorder: A Randomized Clinical Trial. JAMA Netw Open. 2024;7(9):e2435206. PMC11428014.
13. Sandoz Inc. and Pear Therapeutics. FDA Clearance of reSET-O for Opioid Use Disorder. December 10, 2018. NIDA-sponsored randomized controlled trial of 170 patients with OUD over 12 weeks. FDA De Novo Summary DEN160018, accessdata.fda.gov/cdrh_docs/reviews/DEN160018.pdf.
14. Velez FF, Colman S, Kauffman L, Ruetsch C, Anastassopoulos K. Reduced Healthcare Resource Utilization in Patients with Opioid Use Disorder in the 12 Months After Initiation of a Prescription Digital Therapeutic. Adv Ther. 2022;39(9):4131–4145. PMC9402736.
15. Maricich YA, Gerwien R, Kuo A, Malone DC, Velez FF. Real-World Use and Clinical Outcomes After 24 Weeks of Treatment with a Prescription Digital Therapeutic for Opioid Use Disorder. Hosp Pract. 2021;49(5):348–355. doi:10.1080/21548331.2021.1974243.
16. Higgins ST, Kurti AN, Davis DR. Voucher-Based Contingency Management Is Efficacious But Underutilized in Treating Addictions. Perspect Behav Sci. 2019;42(3):501–524.

17. Bolívar HA, Klemperer EM, Coleman SRM, DeSarno M, Skelly JM, Higgins ST. Contingency Management for Patients Receiving Medication for Opioid Use Disorder: A Systematic Review and Meta-analysis. JAMA Psychiatry. 2021;78(10):1092–1102. doi:10.1001/jamapsychiatry.2021.1969. PMC8340014. (74 RCTs, 10,444 participants.)
18. Ginley MK, Pfund RA, Rash CJ, Zajac K. Long-Term Efficacy of Contingency Management Treatment Based on Objective Indicators of Abstinence from Illicit Substance Use Up to 1 Year Following Treatment: A Meta-Analysis. J Consult Clin Psychol. 2021;89(1):58–71.
19. DeFulio A, Furgeson J, Brown HD, et al. Smartphone App-Based Contingency Management and Opioid Use Disorder Treatment Outcomes. JAMA Netw Open. 2024;7(12):e2448167. (Retrospective cohort, 600 patients, WEconnect platform.)
20. U.S. Department of Health and Human Services, Office of Inspector General. Advisory Opinion No. 22-04. Digital Contingency Management Program. 2022.
21. Smith JE, Meyers RJ. Motivating Substance Abusers to Enter Treatment: Working with Family Members. Guilford Press; 2004.
22. Meyers RJ, Miller WR, Hill DE, Tonigan JS. Community Reinforcement and Family Training (CRAFT): Engaging Unmotivated Drug Users in Treatment. J Subst Abuse. 1999;10(3):291–308. doi:10.1016/S0899-3289(99)00003-6.
23. Meyers RJ, Miller WR, Smith JE, Tonigan JS. A Randomized Trial of Two Methods for Engaging Treatment-Refusing Drug Users Through Concerned Significant Others. J Consult Clin Psychol. 2002;70(5):1182–1185.
24. Roozen HG, de Waart R, van der Kroft P. Community Reinforcement and Family Training: An Effective Option to Engage Treatment-Resistant Substance-Abusing Individuals in Treatment. Addiction. 2010;105(10):1729–1738.
25. Brigham GS, Slesnick N, Winhusen TM, Lewis DF, Guo X, Somoza E. A Randomized Pilot Clinical Trial to Evaluate the Efficacy of Community Reinforcement and Family Training for Treatment Retention (CRAFT-T) for Improving Outcomes for Patients Completing Opioid Detoxification. Drug Alcohol Depend. 2014;138:240–243. doi:10.1016/j.drugalcdep.2014.02.013.
26. Dutcher LW, Anderson R, Moore M, et al. The Community Reinforcement and Family Training (CRAFT) Approach: An Effectiveness Study. J Behav Anal Health Sports Fitness Med. 2009;2(1):80–93.
27. Campbell ANC, Nunes EV, Matthews AG, et al. Internet-Delivered Treatment for Substance Abuse: A Multisite Randomized Controlled Trial. Am J Psychiatry. 2014;171(6):683–690. (NIDA Clinical Trials Network, 507 patients across 10 community programs, Therapeutic Education System.)
28. Christensen DR, Landes RD, Jackson L, et al. Adding an Internet-Delivered Treatment to an Efficacious Treatment Package for Opioid Dependence. J Consult Clin Psychol. 2014;82(6):964–972. doi:10.1037/a0037496.
29. Lechien JR, et al. Speech and Voice Quality as Digital Biomarkers in Depression: A Systematic Review. J Voice. 2025. (12 studies, 16,872 participants, AUC range 0.71–0.93.)
30. Schultebraucks K, Wen T, Kronish IM, Willey J, Chang BP. The Voice of Depression: Speech Features as Biomarkers for Major Depressive Disorder. BMC Psychiatry. 2024;24:794. doi:10.1186/s12888-024-06253-6.
31. Larsen E, Murton OM, Song I, et al. Validation of a Mental Fitness Vocal Biomarker for Depression and Anxiety Symptoms. Front Psychiatry. 2024;15:1342835. doi:10.3389/fpsyt.2024.1342835.
32. Multimodal benchmark study of smartphone speech for depression monitoring. NPJ Digit Med. 2025. (3,151 weekly voice diaries, 284 German-speaking adults.)
33. Centers for Medicare & Medicaid Services. Calendar Year 2022 Medicare Physician Fee Schedule Final Rule. Remote Therapeutic Monitoring CPT Codes 98975–98981. Federal Register, November 2021.
34. U.S. Department of Health and Human Services, Office of the Surgeon General. Facing Addiction in America: The Surgeon General's Report on Alcohol, Drugs, and Health. Washington, DC: HHS; 2016.

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